Blessed G/Michael Catholic Secondary School

G-12 Physics Test **Topic: unit 3 Fluid Mechanics** **Date: 07/04/2018 EC**

Choose the best answer from the given alternatives. **Time allowed: 1:00 hr**

A 1. Barometer uses the principle of fluid at rest or pressure gradient for its pressure calculation.
A) True B) False

A 2. As we go upwards, at height there is slight decrease in pressure variation. A) True B) False

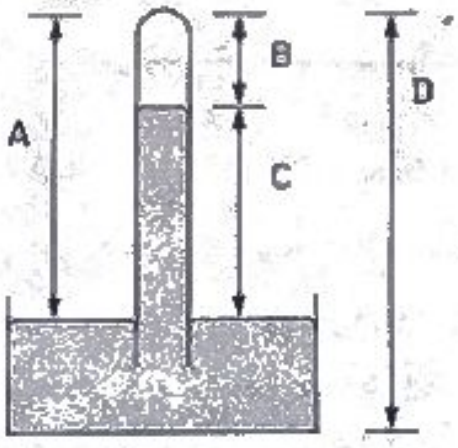
B 3. Manometers are the pressure measuring devices which use the principle of dynamic pressure to measure the pressure difference. A) True B) False

D 4: Which of the following **does not** affect the pressure beneath the surface of a liquid?
A. depth of the liquid ✓ C. strength of the gravitational field ✓
B. density of the liquid ✓ D. area of the liquid surface ×

C 5. A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 50 cm and the pressure inside the cylinder is P. The piston is pushed in a distance of 30 cm, so that the length of the gas column is now 20 cm. The temperature of the gas does not change. What is the new pressure of the gas?
A. 0.6 p B. 0.4 p C. 2.5 p D. 1.5 p

C 6. Which would be the least likely to sink into soft ground?
A. an empty lorry with six wheels × C. a loaded lorry with four wheels
B. an empty lorry with four wheels × D. a loaded lorry with six wheels ✓

C 7. The diagram shows a simple mercury barometer. Which height is a measure of the atmospheric pressure?

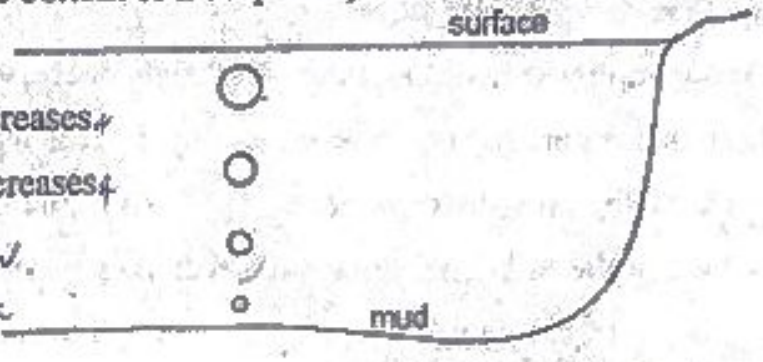


B 8. The bottom of a flat-bottomed aluminum boat has area = 4.0 m^2 and mass = 60 kg. If two fishermen and their fishing gear with total mass of 300 kg are placed in the boat, how much lower will the boat ride in the water? (H_2O density = 1.0×10^3 kg/m^3)
A. 0.15 m B. 0.090 m C. 0.075 m D. 0.060 m

C 9. If the pressure exerted on a floor by a box is 200 kPa, find the weight of the box, given that the bottom of the box has a contact area of 20 cm^2.

A. 600 N B. 300 N C. 400 N D. 3000 N

C 10. Bubbles of gas, escaping from the mud at the bottom of a deep lake, rise to the surface. As the bubbles rise, they get larger. Why is this?



A. Atmospheric pressure on the bubbles increases.

B. Atmospheric pressure on the bubbles decreases.

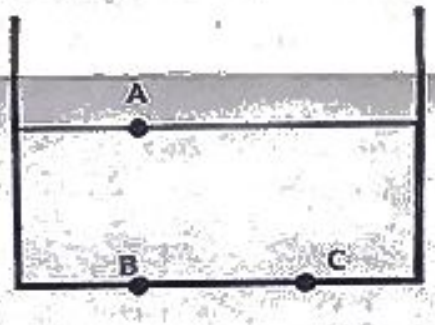
C. Water pressure on the bubbles decreases.

D. Water pressure on the bubbles increases.

B 11. A solid object is made of two materials, one material having density of 1 000 kg/m^3 and the other having density of 4 000 kg/m^3. If the object contains equal masses of the materials, what is its average density?

A. 3000 kg/m^3 B. 1600 kg/m^3 C. 2500 kg/m^3 D. 5000 kg/m^3

D 12. A beaker half-filled with water is exposed to the atmosphere. If the pressure at points A, B and C as shown are P_a, P_b and P_c respectively, which one of the following will be the relation connecting the three?



A. $Pa > Pb = Pc$

B. $Pa > Pb > Pc$

C. $Pa < Pb < Pc$

D. $Pa < Pb = Pc$

B 13. A block of wood has density 0.50 g/cm^3 and mass 1 500 g. It floats in a container of oil (the oil's density is 0.75 g/cm^3). What volume of oil does the wood displace?

A. 3 000 cm^3 B. 2 000 cm^3 C. 1 500 cm^3 D. 1 000 cm^3

B 14. A rigid metal object is dropped into a lake and sinks to the bottom. The density of the water in this lake is the same every-where. As the object sinks deeper and deeper below the surface, what will be the buoyant force on it

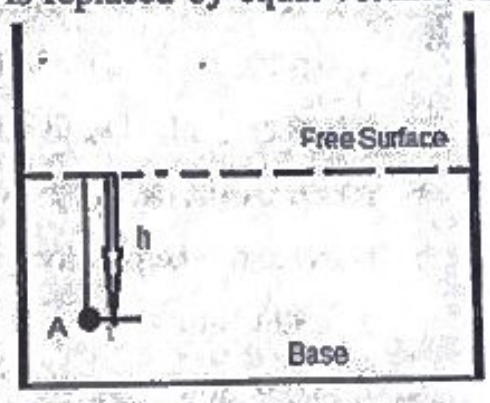
A. It gets greater and greater. C. It gets less and less.

B. It does not change. D. It is equal to the weight of this object.

B 15. A spherical object has a density If it is compressed under high pressure to half of its original diameter, its density will now be

A. $\rho/8$. B. 8ρ. C. $\rho/4$. D. 4ρ. E. 2ρ.

B 16. A beaker is filled with a liquid of density ρ_1 up to a certain height. A is a point, h meter downwards from the free surface such that the pressure at A is P. If the liquid is replaced by equal volume of another liquid of density ρ_2, at what distance from the free surface will the pressure be P now?



A. h C. $\frac{\rho_1+\rho_2}{2}h$

B. $\frac{\rho_1}{\rho_2}h$ D. $\frac{\rho_2}{\rho_1}h$

17. Legend says that Archimedes, in determining whether or not the king's crown was made of pure gold, measured its volume by the water displacement method. If the density of gold is 19.3 g/cm^3, and the crown's mass is 600 g, what volume would be necessary to prove that it is pure gold?

A. 31.1 cm^3 B. 114×10^3 cm^3 C. 22.8×10^3 cm^3 D. 1.81×10^{-2} cm^3

A 18. A Boeing-737 airliner has a mass of 20 000 kg. The total area of the wings is 100 m^2. What must be the pressure difference between the top and bottom of the wings to keep the airplane up?

A. 2000 Pa B. 4000 Pa C. 8000 Pa D. 16000 Pa

B 19. A 20 000-N car on a hydraulic lift rests on a cylinder with a piston of radius 0.20 m. If a connecting cylinder with a piston of 0.040-m radius is driven by compressed air, what force must be applied to this smaller piston in order to lift the car?

A. 600 N B. 800 N C. 1 200 N D. 5 000 N

D 20. A horizontal cylindrical pipe has a part with a diameter half that of the rest of the pipe. If v is the speed of the fluid in the wider section of pipe, then what is the speed of the fluid in the narrower section?

A. v/4 B. v/2 C. 2v D. 4v

C 21. The barometric pressure at sea level is 760 mm of Mercury while that on a mountain top is 715 mm. If the density of air is assumed constant at 1.2 kg/m^3 and g = 10 m/s^2, what is the elevation of the mountain top?

A. 1.25 km B. 1.0 km C. 0.5 km D. 0. 25 m

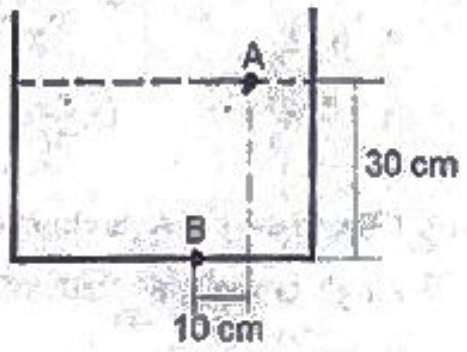
C 22. A horizontal pipe with water flowing through it has a circular cross section that varies in diameter. The diameter at the wide section is 2 times that of the diameter at the narrow section. If the rate of flow of the water in the narrow section is 4.0 L/min, what is the rate of flow of the water in the wide section?

A. 1.0 L/min B. 2.0 L/min C. 4.0 L/min D. 16 L/min

23. A box with a mass of 5 kg, whose bottom measures 10 cm x 10 cm, sits on a table. What is the pressure the box exerts on the tabletop?

A. 0.5 Pa B. 50 Pa C. 500 Pa D. 5000 Pa

24. A beaker is filled with a liquid of specific gravity SG = 0.5 as shown below. What will be the pressure difference between the two points A and B, 30 cm below and 10 cm to the right of point A?



A. 4500 N/m^2

B. 3500 N/m^2

C. 2500 N/m^2

D. 1500 N/m^2

25. If fluid is at rest in a container of a narrow mouth at a certain column height and same fluid is at rest at same column height in a container having broad mouth, will the pressure be different at equal depth from fluid surface?

A. Pressure will be more for narrower mouth C. Pressure will be same for both

B. Pressure will be less for narrower mouth D. None of the mentioned

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1. A	10. C	19. B
2. A	11. B	20. D
3. B	12. D	21. C
4. D	13. B	22. C
5. C	14. B	23. DB
6. CA	15. B	24. D
7. C	16. B	25. C
8. B	17. DA	
9. C	18. AB	